STRUCTURAL BIOLOGY & PROTEIN ENGINEERING LAB

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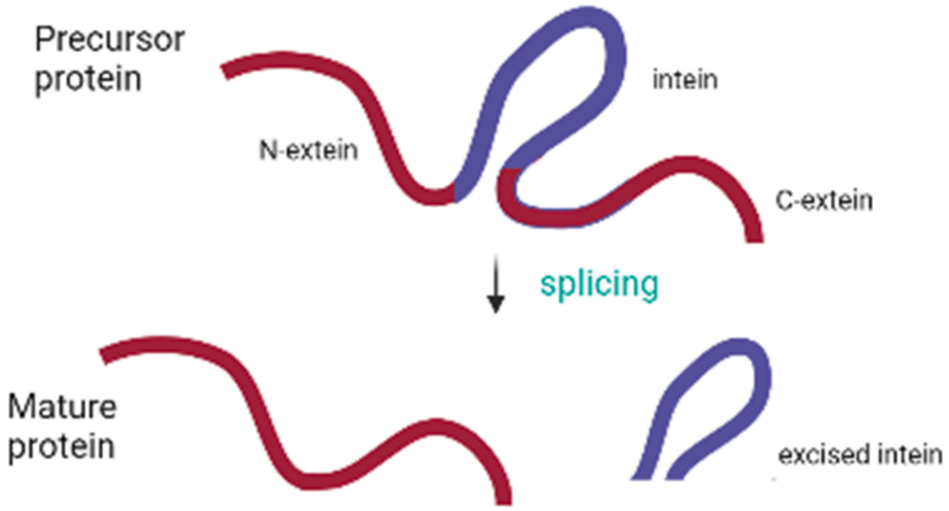
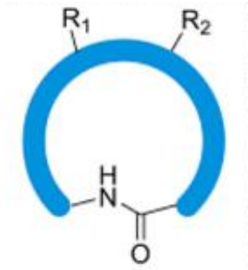
22CH30028

Expression of novel cyclic peptides with inhibitory function in E. Coli.

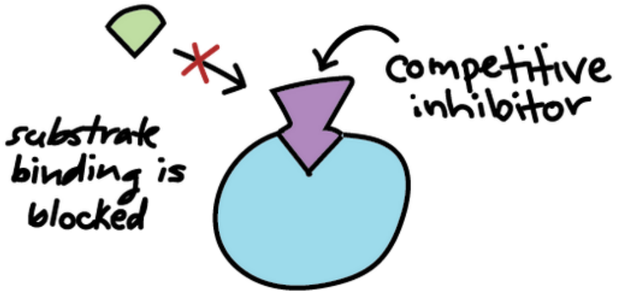
MECHANISM OF ACTION



Intein forms a cyclic peptide



Cyclic peptide inhibits protein-protein interaction



Split Intein which undergoes self-splicing and inhibits a protein signalling network involved in leukemia

PROCEDURE

Design of peptide

DNA amplification → Restriction digestion → Ligation & Transformation → Cell harvest & lysis → Purification

WORKFLOW

CLONING

Gene of interest Inserted into **plasmid** → DNA amplification (PCR) → Restriction digestion (pET 28A vector w/ His Tag) → Ligation (T4 DNA Ligase)

EXPRESSION

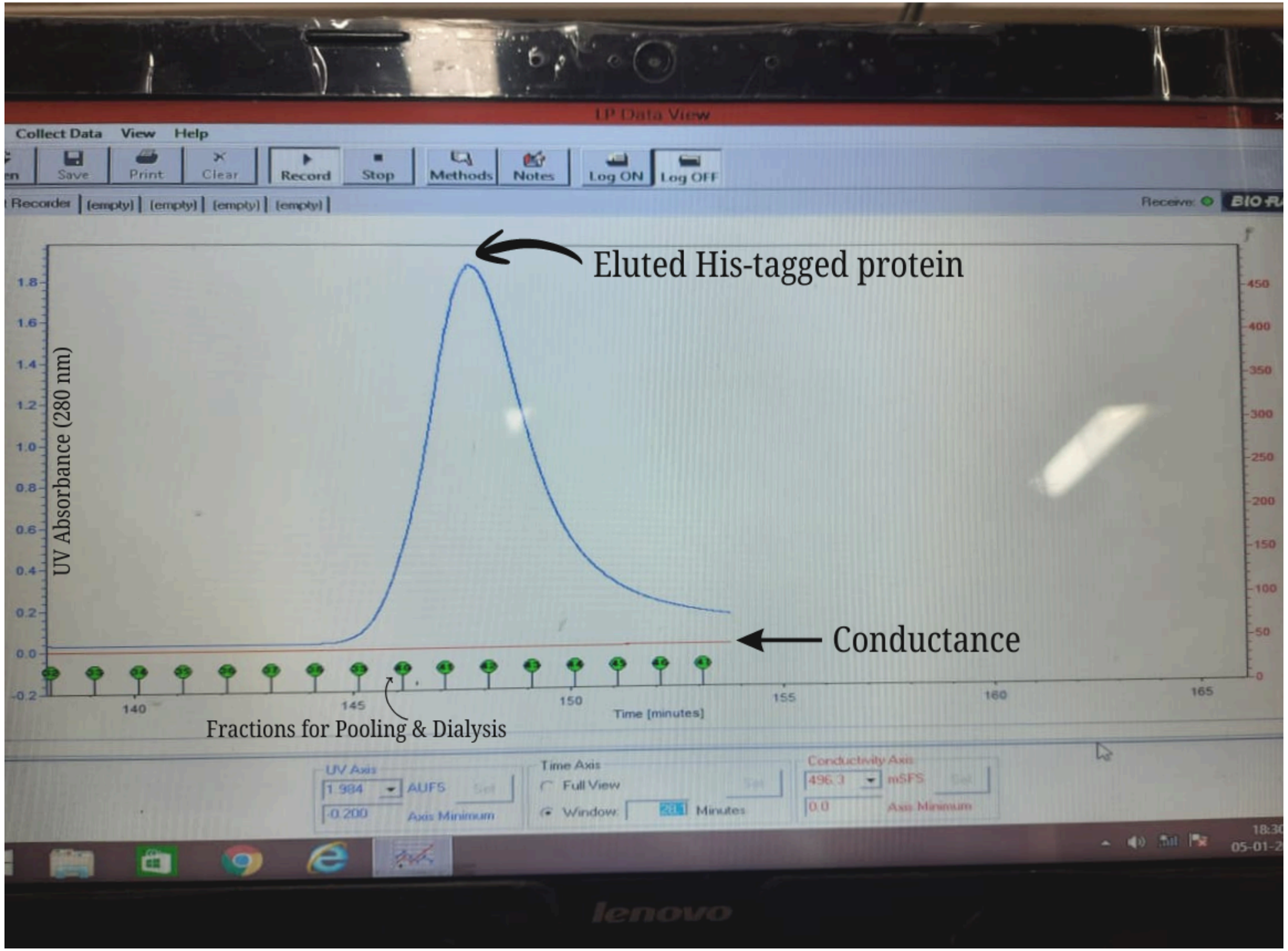
Transformation (uptake by competent cells **DH5α strain**) → Extract plasmid (Miniprep) → Inocculation (Primary Culture) → Induction (with IPTG) (37 degC @ 4 hrs) **[BL21 strain]** → Centrifugation & Sonication

DOWNSTREAM PROCESSING

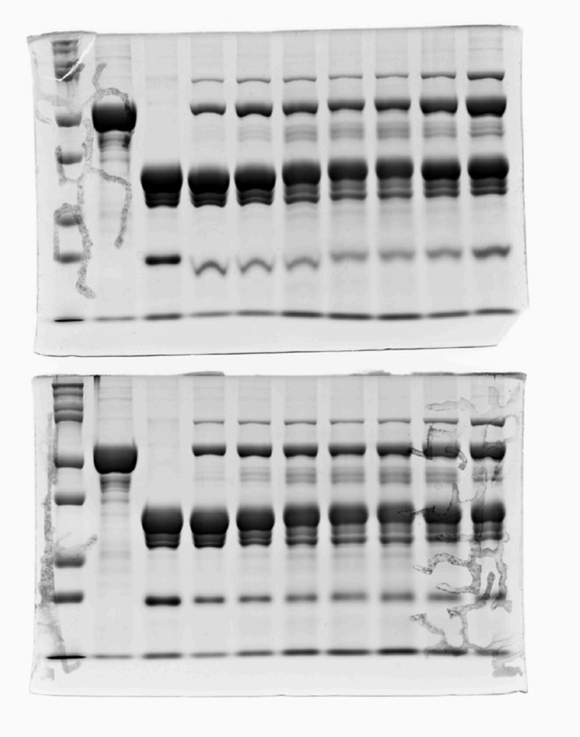
Filtration (Ni-NTA affinity chromatography) → Loading of His-tagged protein → Wash buffer → Elution (with Imidazole) → Dialysis

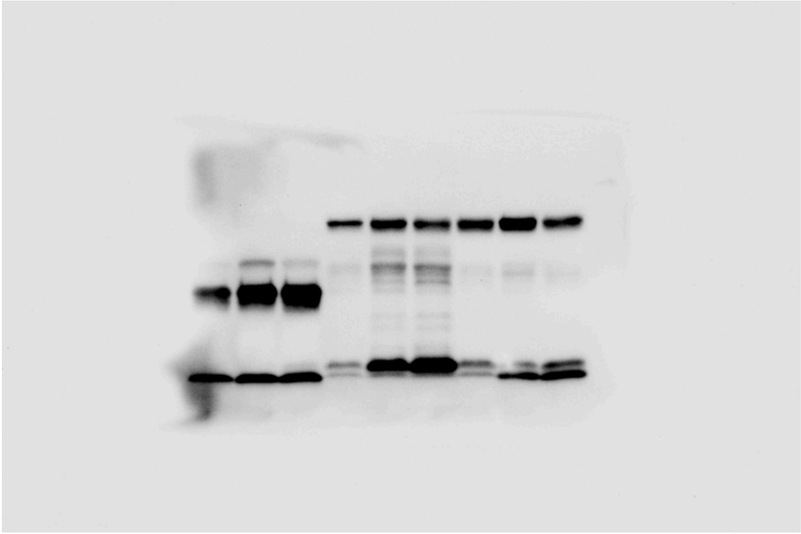
RESULTS

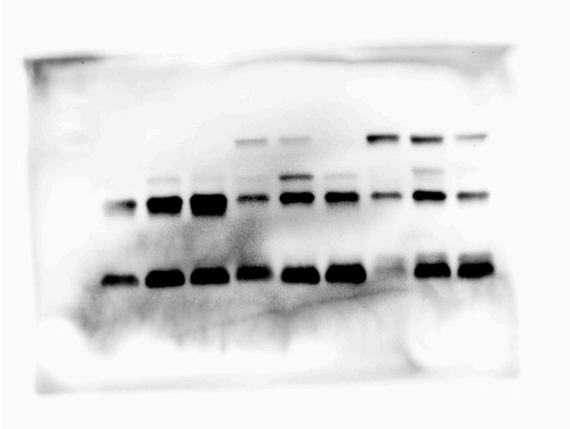
ELUTION with Ni-NTA chromatography



SDS PAGE ANALYSIS







THANK YOU!

* Lab dinner, Dec, 2023